

Background

- Hydrogen is colorless, odorless, and highly flammable, making leaks hard to detect, especially in enclosed public spaces like at the NREL Energy Systems Integration Facility (ESIF). Accurate modeling helps identify high-risk zones, optimize sensor placement, and support emergency response strategies
- The detection system in NREL's hydrogen laboratories is being upgraded for safety. To address this, dispersion modeling is being implemented to quantify hydrogen behavior and inform optimal sensor placement within indoor lab environments
- Hydrogen dispersion modeling combined with machine learning models are being developed to detect leaks quickly and reliably under real-world conditions, with help from machine learning models like the Artificial Neural Network (ANN) and K-Nearest Neighbor with Dynamic Time Warping (K-DTW)

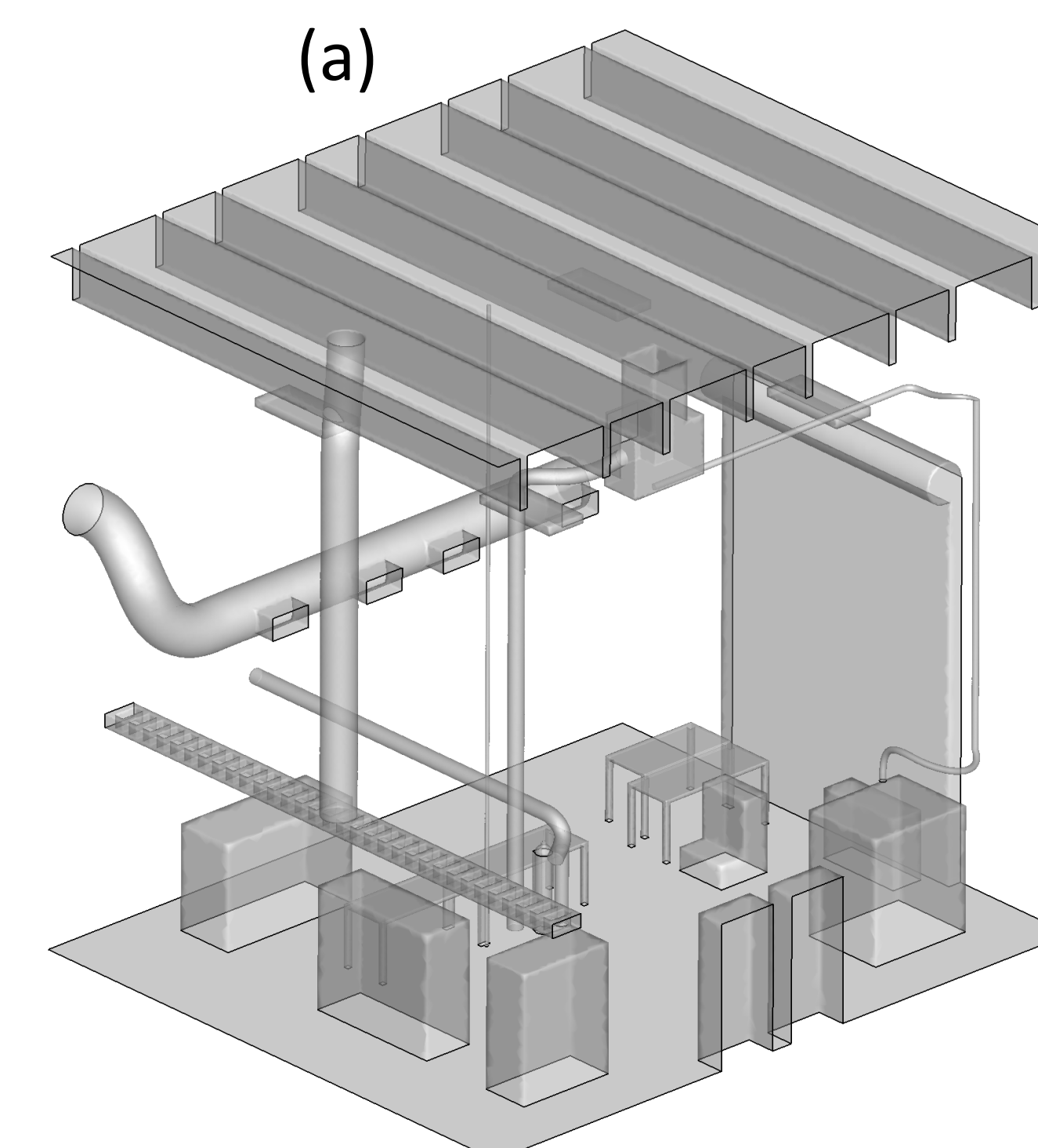


Figure 1. Experimental setup for hydrogen leak detection at ESIF-317 sensor laboratory

Data Preparation

16
sensors

2000 seconds
from each
experiment
and simulation

11 set of
scenarios

- Future model will incorporate and add more leak locations, various size of leak and various pressures of leak

K-Nearest Neighbor with Dynamic Time Warping (K-DTW)

$$D(i, j) = |ts1[i - 1] - ts2[j - 1]| + \min\{D(i - 1, j), D(i, j - 1), D(i - 1, j - 1)\}$$

Goal

- Align multiple sensor data with Dynamic Time Warping to account for changes in speed, phase, or length and get a useful distance metric
- Use the DTW distances to determine the closest training scenarios, then use the K-Nearest coordinate average to get the continuous output
- Estimate leak coordinates by averaging the top-K neighbor locations and validate accuracy through cross-validation, and report error

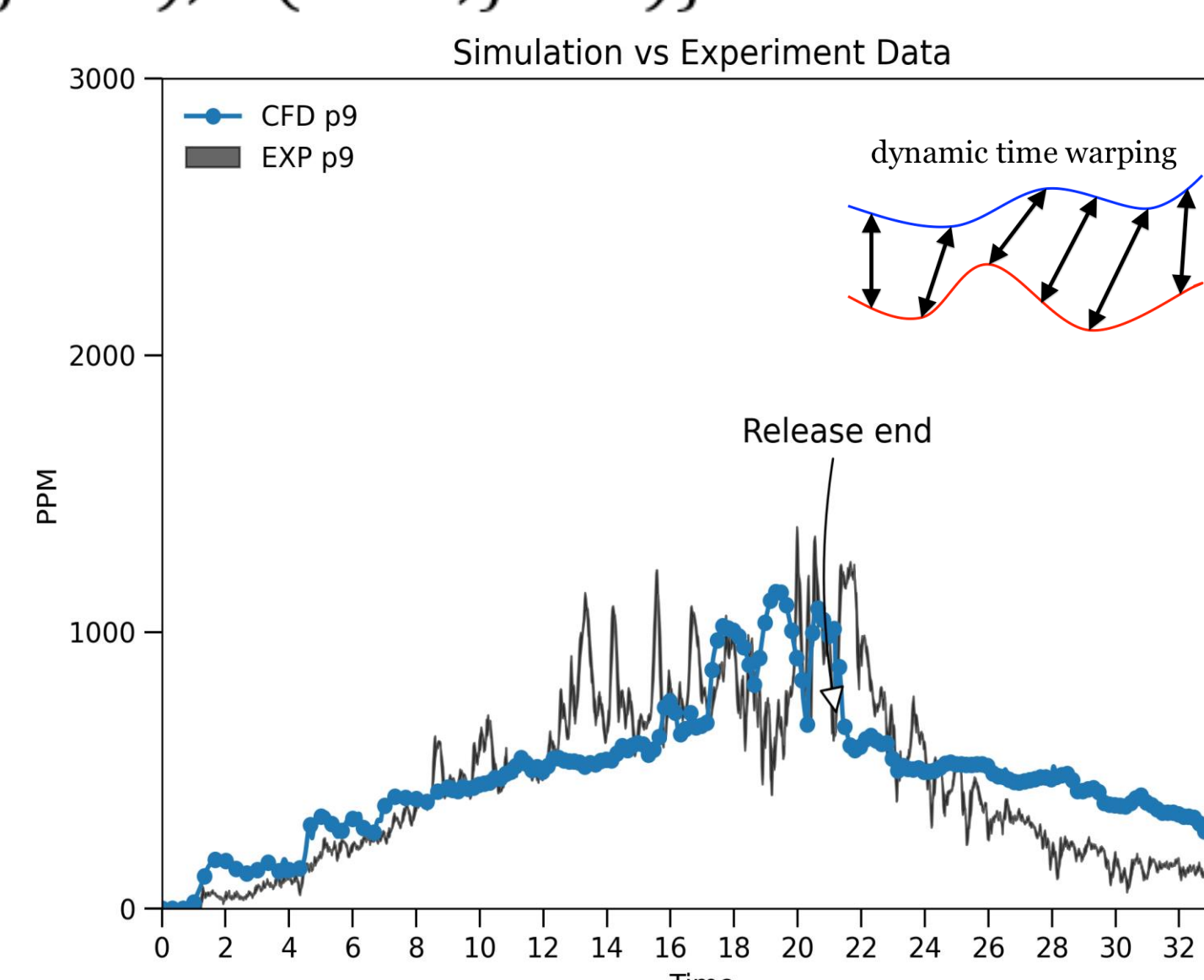
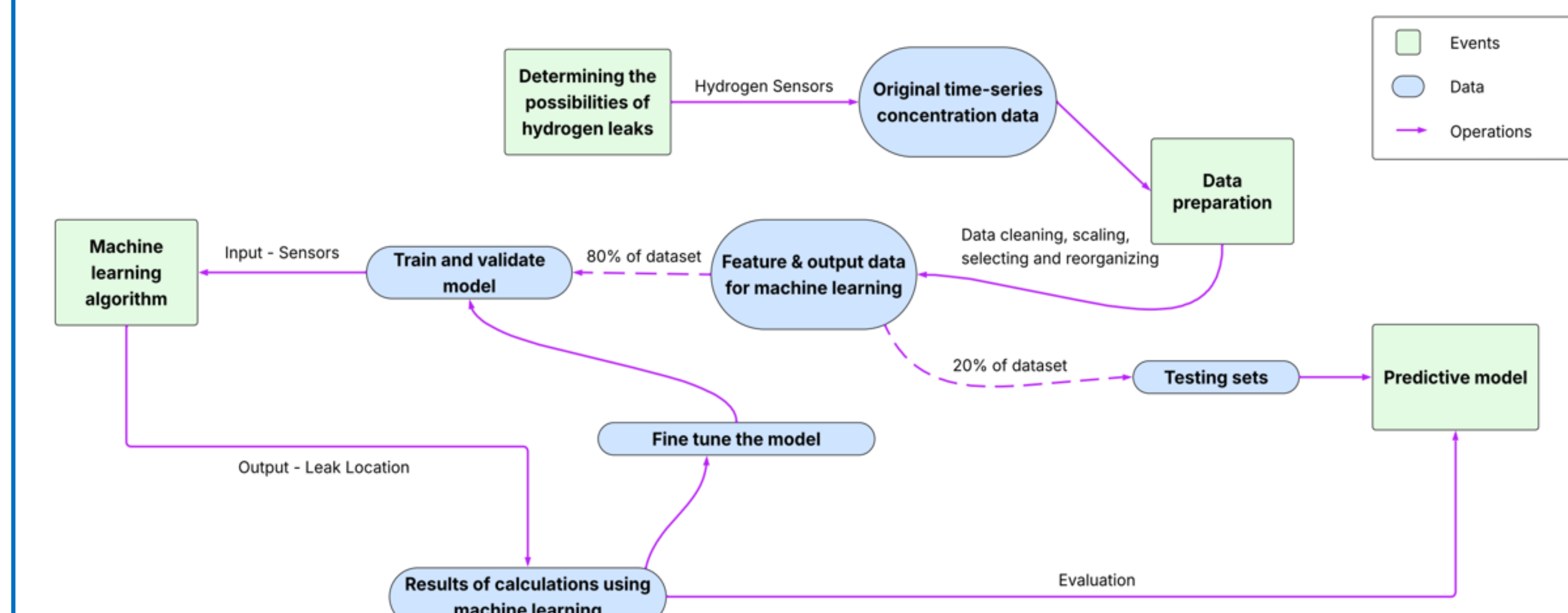


Figure 2. Representative leak time-history

Artificial Neural Network (ANN)

$$y_j^{(k)} = f\left(\sum_{i=1}^n u_{i,j}^{(k)} \times y_i^{(k-1)} + b_j^{(k)}\right) \xrightarrow{\text{Mean Squared Error}} \text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_{\text{true},i} - y_{\text{pred},i})^2 \xrightarrow{\text{Penalty-Aware Loss Function (MSE + Euclidean Penalty)}} \text{Penalty} = \|y_{\text{pred}} - y_{\text{true}}\|_2 = \text{Loss} = \text{MSE} + \lambda(\text{Penalty})$$

Figure 3. Workflow for training and testing a machine learning model using sensor data



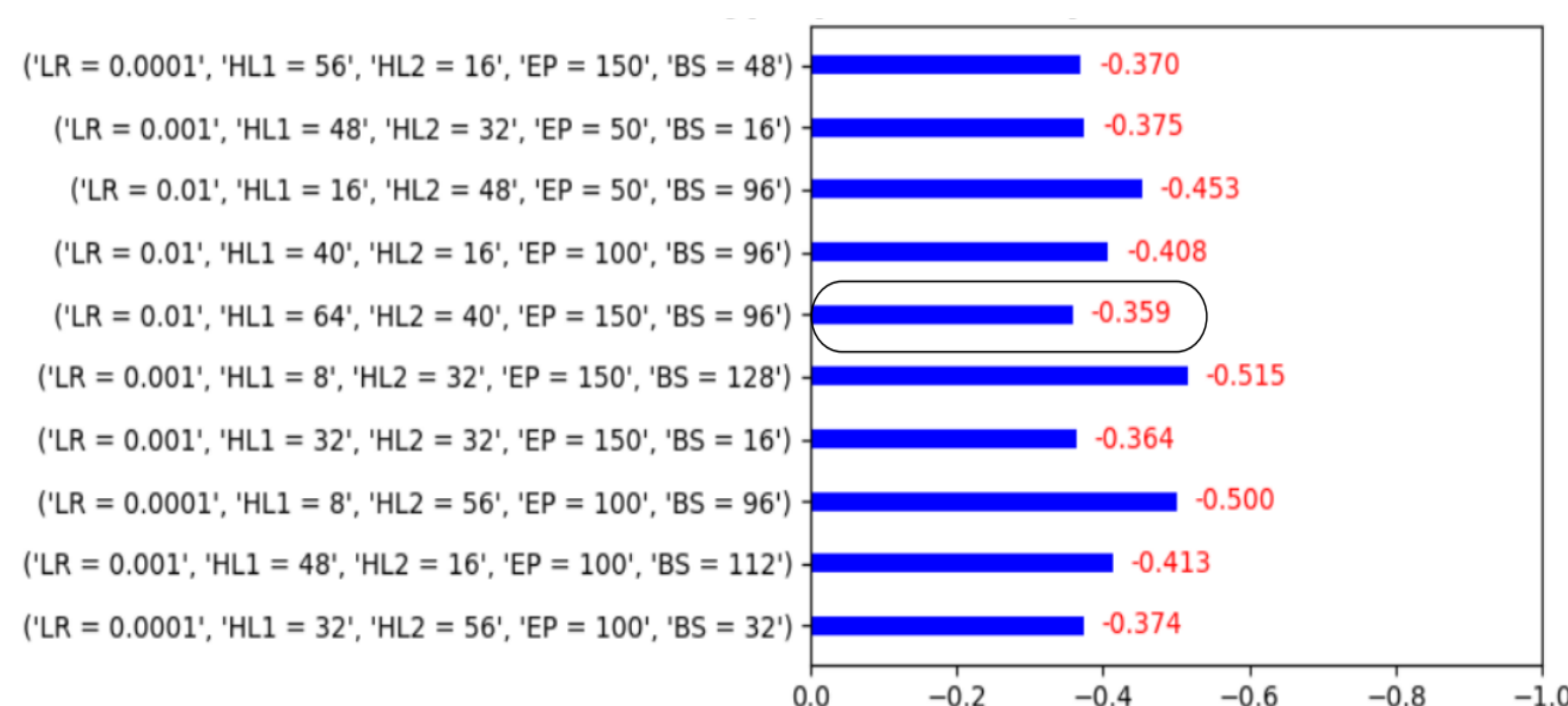
Parameter

- Consists of an input layer with 16 neurons (sensor concentrations), two hidden layers (64 and 40 neurons, with rectified linear unit (ReLU) activation, and an output layer with 2 neurons (predicting X and Z leak coordinates using linear activation)
- Model uses the Adam optimizer with a learning rate of 0.01 and has early stopping to prevent overfitting
- Trained over 150 epochs to predict hydrogen leak locations

Goal

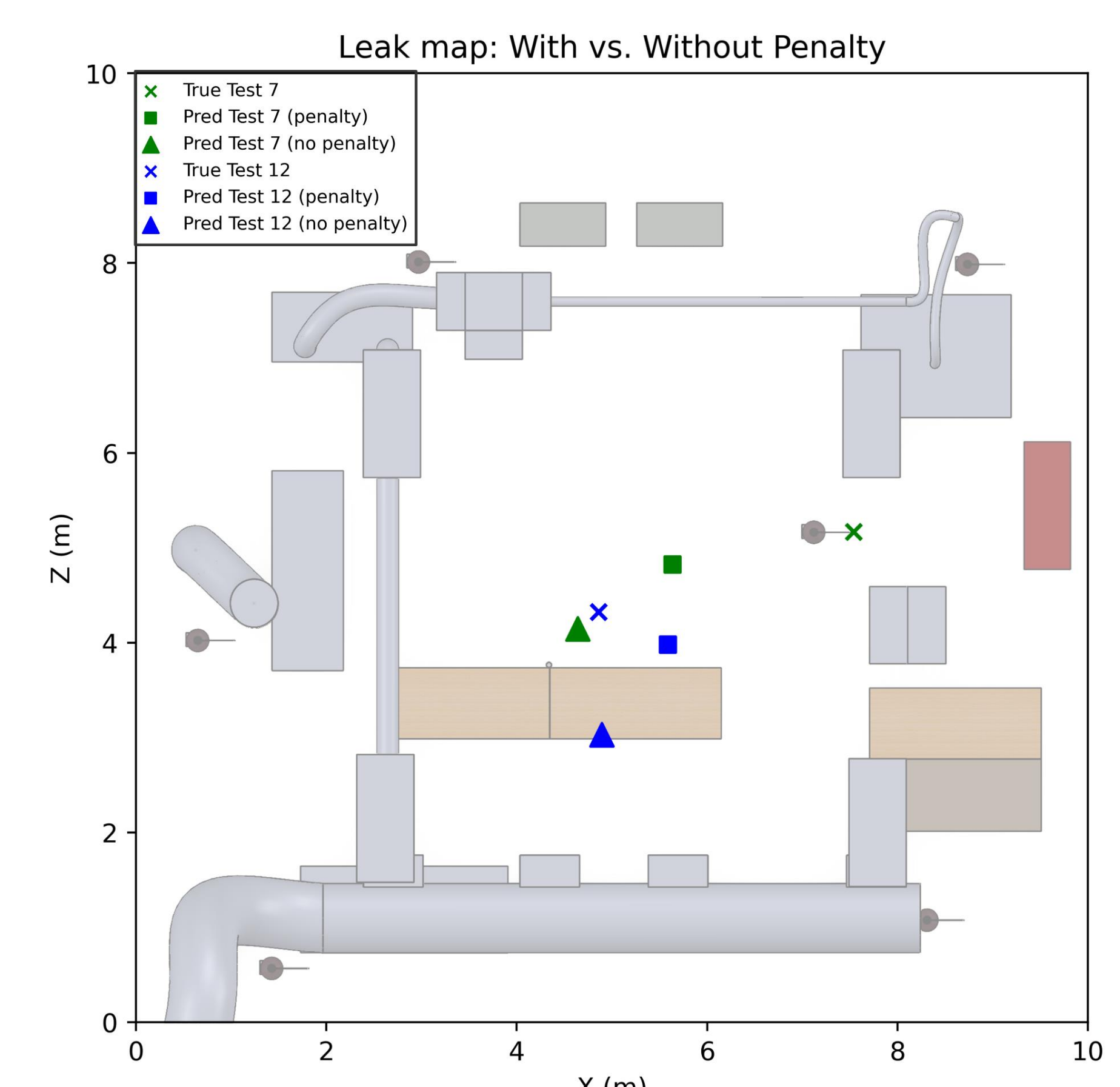
- Used to predict the X and Z coordinates of hydrogen leaks
- Identify patterns in sensor data from 16 gas sensors to estimate leak locations
- Compare ANN models with two different loss functions, one with the Penalty-Aware loss function and the other with only MSE

Hyperparameter Optimization (Mean Test Score)



Results

Figure 4. Leak localization map comparing predicted leak positions using ANN models with and without the Penalty-Aware loss function



- Predictions from the penalty-based model (squares) are visibly closer to the true leak markers ('X'), showing improved localization accuracy with the addition of the penalty term

Model	Test 7 – Distance to Actual Leak in Meters	Test 12 – Distance to Actual Leak in Meters
ANN with Penalty (Optimized)	1.9 m	0.8 m
ANN with No Penalty (Optimized)	3.07 m	1.29 m

Table 1. Statistical comparison of prediction accuracy between optimized ANN models

Conclusion & Future Work

- Add more data for machine learning models to increase accuracy
- Conduct hyperparameter optimization for K-DTW
- Explore other models for better prediction of leak localization
- Deploy these models in other lab environments, where leak vary significantly from controlled laboratory dynamics and sensor performance conditions

References

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- W. Buttner, D. Peaslee, J. Stewart, L. Martens, K. Hartmann, G. Saur, Z. Ma, J. Gifford, M.P. Shah, I. Palin, and O. Robinson, "NREL Hydrogen Sensor Testing Laboratory," in Present. 2024 US Dep. Energy DOE Hydrog. Program Annu. Merit Rev. Peer Eval. Meet. AMR 6-9 May 2025 Arlington, Va., (US DOE, 2024).

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